Linear Mixed-Effects Regression

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1. Data preparation

The longitudinal data are from an experiment to study the effect of intermittent fasting on the weights of 15 participants, each measured weekly during the intervention.

```
df <- dat <- read.csv("~/Documents/IF/dataset/IFStudy-Weights_DATA_2023-05-25_0957.csv")
str(df)
```

```
## 'data.frame':    324 obs. of  6 variables:
##  $ participant_id          : chr  "IF-002" "IF-002" "IF-002" "IF-002" ...
##  $ redcap_event_name       : chr  "registration_arm_1" "preintervention_arm_1" "intervention_arm_1" "postintervention_arm_1" ...
##  $ redcap_repeat_instrument: chr  "" "" "" "" ...
##  $ redcap_repeat_instance  : int  NA NA NA NA NA 1 2 3 4 5 ...
##  $ weekly_journal          : int  NA NA NA NA NA 1 2 3 4 5 ...
##  $ sunday_weight           : chr  "" "" "" "" ...
```

```
dat$sunday_weight <- as.numeric(dat$sunday_weight)
dat <- na.omit(dat) #exclude rows that have missing values (blank, "N/A", NA) in any column
dat <- dat[,c(1,5,6)]
colnames(dat) <- c("ID", "Week", "Weight")
dat$ID <- factor(dat$ID)
table(dat$ID) #number of observations for each subject
```

```
##
## IF-002 IF-004 IF-005 IF-006 IF-007 IF-009 IF-010 IF-012 IF-013 IF-014 IF-017
##     12     13     12     13      3     14     12     10     11      2     25
## IF-018 IF-019 IF-020 IF-021 IF-022
##     13     23     24     21     25
```

```
dat <- droplevels(subset(dat, !(ID %in% c("IF-007","IF-014")))) #exclude insufficient data
dat$Week <- dat$Week - 1 #start from Week 0
str(dat)
```

```
## 'data.frame':    228 obs. of  3 variables:
##  $ ID    : Factor w/ 14 levels "IF-002","IF-004",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ Week  : num  0 2 3 4 5 6 7 8 9 10 ...
##  $ Weight: num  172 168 170 170 168 ...
```

2. Modeling

In a linear mixed-effects regression, each subject has its intercept and slope (i.e., random intercepts and random slopes).

```
fit <- nlme::lme(fixed=Weight ~ Week,
                 random=~Week | ID,      #random intercept and slope for each subject
                 data=dat, method="ML")
summary(fit)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   980.1353 1000.711 -484.0676
##
## Random effects:
##  Formula: ~Week | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev     Corr
## (Intercept) 32.1124424 (Intr)
## Week         0.4340361 -0.263
## Residual     1.3752011
##
## Fixed effects:  Weight ~ Week
##                 Value Std.Error  DF   t-value p-value
## (Intercept) 192.92421  8.622254 213 22.375149  0.0000
## Week         -0.38742  0.118913 213 -3.257988  0.0013
##  Correlation:
##      (Intr)
## Week -0.261
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -2.4548417 -0.5978949 -0.1243981  0.5202517  2.7403246
##
## Number of Observations: 228
## Number of Groups: 14
```

The p-value of .0013 for the random slopes is statistically significant at an alpha level of .01 ($t = -3.257988$, $df = 213$). Hence, reject the null hypothesis that the `Week` effect is zero.

```
# -0.38742 + c(-1,1)* 1.96 * 0.118913
nlme::intervals(fit) #95% confidence intervals for the fixed effects
```

```
## Approximate 95% confidence intervals
##
##  Fixed effects:
##                    lower        est.       upper
## (Intercept) 176.0030431 192.9242104 209.8453778
## Week         -0.6207851  -0.3874179  -0.1540507
##
##  Random Effects:
```

```
##   Level: ID
##                             lower       est.      upper
## sd((Intercept))       22.1696586 32.1124424 46.5144266
## sd(Week)               0.2940263  0.4340361  0.6407159
## cor((Intercept),Week) -0.6662934 -0.2632715  0.2587859
##
##  Within-group standard error:
##    lower     est.    upper
## 1.246746 1.375201 1.516891
```

```
coef(fit)
```

```
##        (Intercept)       Week
## IF-002    170.4148 -0.2407723
## IF-004    209.1098 -0.2623625
## IF-005    193.1003 -0.1691887
## IF-006    193.9551 -0.7744081
## IF-009    187.8938 -1.4995722
## IF-010    156.4389  0.4842542
## IF-012    225.4052 -0.2840748
## IF-013    266.4440 -0.3988047
## IF-017    174.2315 -0.5072223
## IF-018    221.0348 -0.7563314
## IF-019    144.2561 -0.1224793
## IF-020    171.9099 -0.1251854
## IF-021    224.1454 -0.3650981
## IF-022    162.5992 -0.4026047
```

3. Summary

A one unit increase in time (a week) is associated with an average weight **loss** of 0.39 lbs (95% confidence interval 0.15-0.62 lbs), according to a longitudinal study of 14 subjects over a minimum of 12 weeks and a maximum of 26 weeks from 2021 to 2022.

When examining the individual fit, IF-010, on the contrary, was gaining weight during the intervention.